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The impact of mine ownership on trade of metal ores

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ABSTRACT

Metals are essential to the global economy, yet traditional criticality assessments, often based solely on the geographic concentration of mining production, overlook the corporate control dimension of risk. Here, we analyzed whether ownership structures affect trade patterns in critical minerals and examined how production and corporate control diverged from 2000 to 2022. We developed a comprehensive country-level dataset using S&P Capital IQ Pro for 12 key metals and metal ores, calculated Herfindahl–Hirschman indices (HHI) for production and ownership, and statistically tested the relationship between foreign mine ownership and international trade flows using logistic and fixed-effects regressions. Finally, we built scenarios for production and ownership in 2030 to match demand estimates from the International Energy Agency (IEA). Results showed only 2%–14% of global ore trade value overlapping with existing foreign ownership ties and no statistically significant relationship between foreign mine ownership and trade flows. Additionally, clear divergences emerged between ownership and production concentration: cobalt production was highly geographically concentrated (HHI of 4602 in 2022) but had dispersed corporate ownership (HHI of 1985), and high-income countries frequently held substantial ownership stakes despite declining shares of actual production. Projected scenarios indicated continued shifts, notably reduced cobalt and lithium production shares in traditional producer countries, offset by growing Canadian and Australian ownership. Although market dynamics do not appear to be influenced by ownership structures today, corporate control remains a potential lever for supply chain disruption. This underscores the need to incorporate ownership into criticality assessments for a more comprehensive understanding of supply risks.

1. Introduction

Metals are critical to the global economy, particularly for the clean energy transition (Graedel et al., 2012; Vidal et al., 2022). As demand continues to grow, ensuring a secure and sustainable supply of these metals has become a central policy priority for many countries (Schrijvers et al., 2020; Mudd et al., 2024; Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs (European Commission) et al., 2023). To this end, countries rely on criticality assessments which have very different methodologies (Hatayama and Tahara, 2018; Helbig et al., 2021; Schrijvers et al., 2020). Criticality here refers to the dependence of a country on metals and its vulnerability to metal supply chains (Hatayama and Tahara, 2018; Schrijvers et al., 2020; Graedel et al., 2012). Many criticality assessments rely on geographic concentration of production as a primary indicator of supply risk, under the assumption that if extraction is concentrated in a limited number of

countries, disruptions in these countries could pose significant threats to supply security (Frenzel et al., 2017; Bucciarelli et al., 2025).

However, geographic concentration alone overlooks a critical dimension: corporate control of mining activities. While the geological distribution of mineral deposits determines where mines are physically located, the control of these mines – that is, where the controlling companies' headquarters are located – remains largely in the traditional economic centers (Ericsson et al., 2024b). Decisions on production levels, investments, and pricing are thus frequently made by companies headquartered far from the sites of extraction, potentially leading to different supply dynamics than those suggested by geographic concentration alone. Foreign direct investment (FDI), investment from companies or individuals based in one country into business interests located in another, illustrates why corporate ownership matters for supply security. FDI often drives strategic decisions that influence the pace and intensity of extraction (Long et al., 2017). FDI can “lock up” resource

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supplies by concentrating control in the hands of foreign investors, thereby amplifying national security risks (Moran, 2014). Conversely, under robust regulatory frameworks, FDI may promote diversification and enhance accountability through market scrutiny (Kotschwar et al., 2012). These dual roles of FDI raise a crucial question: to what extent do trade flows reflect, or diverge from, the underlying patterns of corporate control? For instance, (Sun et al., 2024) examined overseas investments and domestic demand for lithium, nickel, cobalt, and platinum in 2019, finding an overlap between investment partners and trade flows. Without statistical analysis, however, it remained unclear whether this overlap was coincidental or indicative of a systematic pattern in which foreign-owned mines preferentially export metals to their owners' home countries. In Andrieu et al. (2025), we used input–output analysis with mine ownership data to show that foreign ownership had no measurable impact on trade flows, but this analysis was limited to 2022 and a few metals.

Both production- and ownership-based perspectives on supply concentration offer valuable insights. One common metric for quantifying concentration is the Herfindahl–Hirschman index (HHI) (Herfindahl, 1950; Hirschman, 1980). The HHI is calculated by summing the squares of market shares, with lower values indicating a more evenly distributed market and higher values reflecting dominance by a few producers. HHI has been used to approximate the probability of supply disruption (Graedel et al., 2012; Helbig et al., 2016; Nassar et al., 2015). It has been calculated for production at the mine level for cobalt (van den Brink et al., 2020) and also at the ownership company level for cobalt (Ericsson et al., 2024b) but both metrics (mine production and ownership) have not been compared at the national level across several metals.

Furthermore, given that mining projects have long lead times, and ownership structures are often established well in advance of production, it is theoretically possible to project future ownership and production scenarios. Doing so could provide early warnings of potential supply chain vulnerabilities. Despite this possibility, no study has focused on forecasting how both production and ownership might evolve together.

Here, we address these gaps by developing a comprehensive dataset of corporate ownership for global mine production of 12 key metals and metal ores (Uranium, Lithium, Copper, Cobalt, Iron ore, Manganese, Zinc, Nickel, Silver, Gold, Lead, Bauxite) aggregated at the country level from 2000 to 2022. Using this dataset, we calculate and compare HHI trends derived from production and from ownership. Next, we statistically analyze whether ownership structures impact trade flows of these metals. Finally, we construct projections of future production and ownership up to 2030, specifically for lithium, copper, and cobalt, based on anticipated production from existing mining projects. This dataset and our results offer a valuable resource for researchers seeking to incorporate corporate control into raw material criticality assessments.

2. Methods

2.1. Establishing a baseline ownership and production database

Data sources. Reliable mine-level statistics remain scarce. National compilations from the British Geological Survey (BGS) (Idoine et al., 2024) and the USGS (Survey, 2025) offer complete time-series, but only as country totals. Finer spatial detail must therefore come from proprietary platforms such as S&P Capital IQ Pro (formerly SNL) or Rystad Energy. These sources are widely used (Berman et al., 2017; Christensen, 2019) yet still contain gaps: satellite imagery suggests that as many as 50% of mining sites are absent in some jurisdictions (Maus and Werner, 2024). Remote-sensing inventories (e.g. 1 171 sites in Maus and Werner, 2024) and individual company reports (Sun et al., 2024) improve coverage but lack consistent ownership or forward projections. Because our study requires a global, forward-looking panel, we adopt

S&P Capital IQ Pro (download January 2024; 37 400 properties, 7 750 operating mines) as the core dataset and fill residual gaps with BGS values and remote sensing.

Countries are grouped into income classes according to the World Bank's 2022 classifications, which divide economies into four income groups—low, lower-middle, upper-middle, and high. These classifications are based on Gross National Income (GNI) per capita, measured in U.S. dollars using the Atlas method. The Atlas method accounts for economic growth, inflation, exchange rates, and population changes, with thresholds adjusted annually for inflation based on a weighted average of GDP deflators from key economies.

Metal selection and coverage. We selected the 12 metals or metal ores that, in 2022, had at least 50% coverage in S&P compared to BGS country totals: uranium (91%), lithium (90%), copper (86%), cobalt (85%), iron ore (82%), manganese (80%), zinc (75%), nickel (71%), silver (66%), gold (65%), lead (60%), and bauxite (50%), with the remaining coverage being unknown. Data are aggregated by country, and missing values are filled using BGS data where possible. For lithium before 2012, BGS data were not available so only S&P data are retained (i.e., we do not introduce an “unknown” category). USGS data is as good as BGS data but BGS was selected here because it was easier to download.

Defining control. In the mining industry direct ownership, whether through a majority stake or significant partial holdings via joint ventures, is the most common mechanism for exercising control over a mine and its production (Ericsson et al., 2020). Here, we rely on the variable “control share” as computed by S&P, which corresponds to the equity percentage held. If a company has a control share of 50% or more, we consider that it actually controls 100% as it has the majority of votes. Else, we consider an entity to exercise control only if it holds more than 10% equity (Ericsson et al., 2020). The total 100% of control is then disaggregated proportionally among the companies that meet this threshold.

One of the main limitations of our work is that some apparently independent firms in our dataset may be controlled by the same parent group. S&P Capital IQ Pro lists the immediate equity holders of each property. In practice, mining firms often employ multi-layered corporate structures (holding companies, joint-venture vehicles, and shell subsidiaries registered in tax havens) to ring-fence liabilities or optimize taxation. Untangling these hierarchies to the level of ultimate beneficial owners would require forensic matching across company registries, which sits outside our present scope but is an important area for future research.

Another limitation is that our analysis treats all equity holders symmetrically, yet the objectives and constraints of state-owned enterprises (SOEs), publicly listed transnational corporations (TNCs), sovereign-wealth funds, and privately held juniors differ markedly. SOEs often pursue strategic resource security, vertical integration, or domestic-industrial policy alignment rather than purely profit-maximizing goals. By contrast, shareholder-driven majors typically target portfolio risk-return optimization and dividend stability, while exploration-focused juniors seek to monetise deposits via selling stakes to or entering joint ventures with larger partners. These divergent mandates mean that two investors holding identical mine stakes can exert very different influences on production, pricing, and downstream trade.

Concentration metric. The HHI is a widely used measure of market concentration, with higher values indicating that a market is dominated by fewer entities, reflecting a higher level of concentration and less competition (Glöser et al., 2015). We calculate the HHI for each metal using the formula:

$$HHI = \sum_{i=1}^N s_i^2 \times 10\,000,$$

where s_i represents the market share of entity i within the total market.

2.2. Comparison with trade

Two methods to compare trade and control over mines have already been employed in the literature. Sun et al. (2024) utilized their material flow analysis database and identified which of these flows overlapped with mining FDI flows. While some flows overlapped, no statistical analysis was carried out. In Andrieu et al. (2025), we used input–output analysis to map flows from mines to final products. We showed that foreign mine control had no measurable impact on trade flows in 2022.

A full material flow analysis is beyond the scope of this paper, as ownership's impact on trade is most relevant at the first transformation stage (ores and concentrates), where products remain closely linked to mine output and undergo minimal processing. Input–output analysis, while useful, is constrained by database coverage, and the most advanced database for studying metals, GLORIA, lacks specific data on lithium, cobalt, and manganese. Instead, we analyze monetary values from trade data at the ores and concentrates level, as these better reflect metal content than tonnages, which aggregate ores and concentrates with varying metal/ton ratios. We use the BACI trade database, which provides bilateral trade flow data for 200 countries at the product level, covering 5000 products (Gaulier and Zignago, 2010). We used the ores and concentrates product codes for bauxite (260600), cobalt (260500), copper (260300), gold (261690), iron ore (260100), lead (260700), manganese (260200), nickel (260400), silver (261610), uranium (261210), and zinc (260800), and the carbonates code for lithium (283691).

To investigate the extent to which FDI influences patterns of international trade in metal ores and concentrates, we denote the monetary value (in constant 2020 USD) of ores and concentrates exported from producing country i to consuming country j , for metal m , in year t , by V_{ijmt} . Using monetary rather than physical quantities ensures comparability across various ore grades and mineral types, which differ substantially in metal content. We further denote by W_{ijmt} the tonnage of mine production in exporting country i that is controlled by companies headquartered in importing country j . This metric captures the intensity of ownership ties directly linking producing and consuming nations. Most country pairs in our dataset report no trade ($V_{ijmt} = 0$), no foreign ownership ($W_{ijmt} = 0$), or both, resulting in substantial sparsity. We adopt an analytical framework specifically designed to manage these features.

We proceed in three logical steps: first, we quantify the practical overlap between trade and ownership; second, we statistically test whether the presence of ownership relationships predicts the existence of trade ties; third, we assess whether stronger ownership ties predict higher trade intensity when both relationships exist. See the method box 1 in the supplementary information for an overview of the methods.

Overlap of trade and ownership. To assess the importance of foreign mine ownership in global ore trade, we first quantify how much of the observed trade can be matched to existing ownership relationships. For exporter i , total export value of metal m in year t is given by:

$$V_{imt} = \sum_j V_{ijmt}.$$

We assume trade flows could theoretically reflect ownership patterns if consuming countries were inclined to import ores primarily from mines they control abroad. Thus, we distribute country i 's total exports V_{imt} among its trading partners j proportionally to the share of mine production in i owned by firms from j . Specifically, the ownership-weighted export value is computed as:

$$\tilde{V}_{ijmt} = V_{imt} \frac{W_{ijmt}}{\sum_k W_{ikmt}},$$

where the denominator ensures the shares sum to unity across all importing countries k .

However, because actual trade flows V_{ijmt} may not align with these ownership-weighted flows, owing to market forces, existing contracts, or logistical factors, we cap the ownership-matched trade value at the actual observed flow to prevent overstating the overlap:

$$C_{ijmt} = \min(V_{ijmt}, \tilde{V}_{ijmt}).$$

Summing over all dyads yields the global overlap ratio, defined as:

$$\text{OverlapShare}_{mt} = \frac{\sum_{i,j} C_{ijmt}}{\sum_{i,j} V_{ijmt}}.$$

This ratio measures the fraction of observed global trade value that could, in principle, be directly associated with foreign ownership ties, thus providing a descriptive upper bound of ownership-trade alignment. It does not imply causation but quantifies the extent of potential control that ownership relationships may theoretically exert over global trade flows.

Incidence: Does ownership predict trade relationships? Next, we statistically test whether the existence of foreign mine ownership relationships increases the probability of observing an actual bilateral trade flow. We define an indicator variable for trade-ownership overlap:

$$O_{ijmt} = \begin{cases} 1 & \text{if } V_{ijmt} > 0 \text{ and } W_{ijmt} > 0, \\ 0 & \text{otherwise.} \end{cases}$$

To isolate the effect of ownership presence, we employ a logistic regression model with year-fixed effects:

$$\text{logit}[\Pr(O_{ijmt} = 1)] = \alpha + \beta \mathbf{1}\{W_{ijmt} > 0\} + \tau_t + \varepsilon_{ijmt},$$

where $\mathbf{1}\{W_{ijmt} > 0\}$ is an indicator taking the value one if foreign ownership exists and zero otherwise, and τ_t captures global year-to-year shocks influencing all country-pairs equally (such as commodity price cycles or global economic fluctuations). Parameter β is interpreted via its odds ratio e^β , indicating how many times more likely a trade flow is observed when foreign ownership ties exist compared to when they do not. Statistical significance of this parameter (assessed using a Wald test) confirms whether foreign ownership meaningfully enhances the probability of bilateral trade connections.

Intensity: Does greater ownership volume predict greater trade? Finally, we examine the strength of the association between the volume of foreign-controlled production and trade value, conditioned on the existence of both relationships. We restrict analysis exclusively to dyads with observed non-zero trade and ownership ($V_{ijmt} > 0, W_{ijmt} > 0$) to avoid biases from zero-inflated data and to directly measure intensity. Using a two-way fixed-effects regression, we estimate:

$$\ln(V_{ijmt}) = \alpha + \gamma_i + \delta_j + \tau_t + \beta \ln(W_{ijmt}) + \varepsilon_{ijmt},$$

where γ_i and δ_j are exporter- and importer-specific fixed effects capturing all time-invariant country characteristics (such as geology, policy environment, infrastructure), and τ_t again absorbs global annual shocks. We cluster standard errors by exporter-importer pair to account for correlation within dyads across years.

The coefficient β represents the elasticity of trade value with respect to foreign-owned mine production. An elasticity significantly greater than zero indicates that, among country-pairs where trade and ownership coexist, greater foreign-controlled production systematically corresponds to higher trade volumes.

Dynamic within-pair test. To further probe whether changes in mine ownership are followed by changes in trade between the same two countries, we estimated an additional regression. We used Poisson pseudo-maximum likelihood (PPML), which is preferred when many trade flows are zero because it can use those zeros directly without transforming the data. The model relates bilateral trade in ores and concentrates to the previous year's ownership share for the same

exporter–importer pair. We included exporter-by-year and importer-by-year fixed effects, which hold constant all time-varying conditions on each side (such as supply shocks, demand swings, and policy changes). This means the test focuses on year-to-year movements within a given country pair rather than on persistent differences across pairs. We also ran versions that include ownership shares lagged by two and three years to allow for delayed effects. Standard errors were clustered by country pair. Because many country pairs never trade or show little change in ownership over time, we checked whether the estimator could find a stable solution.

2.3 Future production and ownership

The S&P database provides mine ownership data along with information on future mines and projects under construction. While it includes annual production forecasts up to 2040, these projections are incomplete. Many expected future mines lack production estimates. To bridge these gaps, we developed a methodology to supplement missing data, focusing on three key energy transition metals: lithium, cobalt, and copper.

2.3.1 Filling the gaps in 2022

To address data gaps for 2022, we compare country-level production figures from S&P and BGS. If S&P's data exceed BGS's, we retain S&P's values, assuming BGS may have underreported production. When S&P's figures are lower, we estimate missing values. Precise filling of these gaps is not possible; if the data were easily accessible, it would already be included in S&P's database.

We use remote sensing to identify mining locations (Maus and Werner, 2024; Tang and Werner, 2023) and geospatial matching techniques to match them with the S&P database. The remote sensing dataset is detailed in another publication currently in preparation (Barzin, 2026). Typically, remote sensing data captures the surface area associated with mines but cannot distinguish between active and inactive mines, calculate the depth of the mine, or detect underground mining activity, creating challenges when assigning production values. To overcome this, we combine information on the mines' spatial footprints with remotely sensed particulate emissions data (Barzin, 2026). This approach allows us to focus on active mines. By multiplying the mine surface area with the level of airborne particle presence, we obtain a first-order approximation of production volumes, which we use to allocate the missing country-level production among the mines lacking production data. Although many factors (such as geology, efficiency, and mine depth) influence production, using surface area and pollution as proxies is a practical improvement over ignoring the unreported mines altogether. To determine which metals are extracted at each site, we limit our analysis to mines already listed in the S&P database, as it provides that information.

Our gap-filling process follows these steps: collect remote sensing and particulate emissions data, match identified mines with S&P coordinates, exclude inactive mines or those already covered by S&P, and allocate missing production based on surface area and emissions. Gap filling for 2022 is only required for 15% of global cobalt production, 14% of copper production, and 10% of lithium production. This disaggregation is intended as a plausible approximation rather than an exact measure.

2.3.2 Projecting future production

Projecting future production is inherently uncertain, as it depends on economic conditions, geopolitical factors, and the pace at which planned projects transition into producing mines. The most reliable sources for future production estimates are technical reports, feasibility studies, and pre-feasibility studies from exploration or mining companies because these documents follow international standards. While S&P Capital IQ Pro compiles much of this data and provides forecasts

up to 2040, it still contains significant gaps, with many projects lacking start dates and production estimates.

Where available, we use S&P's future production values, as they are based on company reports. For missing data, we estimate production by making assumptions about start dates, closure timelines, and production levels based on reserves or existing mine outputs. We project future production only until 2030 as uncertainties both in terms of future mines and ownership structures become too important after that date.

Step 1. Estimating the start year: Each S&P-listed project is assigned a “mine stage,” but only some include a projected start date. For those without one, we infer a start year based on typical development timelines (see Supplementary Table 1, built after a consultation of mining experts and by one of the authors who has experience in mine engineering). For instance, we assume that exploration-stage mines could open between 11 and 40 years from now. To distribute ownership across all potential projects, we assume that mines begin production at the lower bound of their expected range (e.g., 11 years for exploration-stage projects) with zero initial output, ramping up linearly to full production by the upper bound. This approach is not realistic at the mine level (in reality some projects will start producing at full capacity very quickly while others will never open) but ensures a balanced representation of future ownership distribution.

Step 2. Estimating the closure year: If S&P provides a closure year, we use it. Otherwise, if S&P's production projections end before 2030, we take the last available projection year as the closure year. Otherwise, we assume that existing mines will begin closing after 2030.

Step 3. Estimating future production from reserves or resources: With start and closure years defined, we estimate annual production volumes. Where S&P provides production forecasts, we use them. Otherwise, we rely on empirical relationships between reserves and production. A well-documented log–log relationship between daily metal production and reserves (Taylor, 1986) holds for our three key metals, and is consistent with past studies (Taylor, 1986; Singer et al., 2000; Long, 2009), (see Supplementary Figures 1–3). Although Taylor's original formulation relates reserves to capacity rather than actual production, the ratio of actual production to capacity is relatively stable over time (e.g., approximately 82% for copper; Parpan et al., 2026). Hence, calibrating Taylor's law directly on observed production implicitly captures this stable production-to-capacity ratio. If S&P lists reserves but not future production, we use this relationship to estimate production levels. When reserves data are unavailable, we instead utilize mineral resources estimates, which are more frequently reported. To convert resources to approximate reserves, we apply a linear regression model fitted to mines reporting both reserves and resources (Supplementary Figure 4). The resulting reserve approximations are then used in Taylor's law to estimate production levels.

Since our goal is not to precisely forecast location specific production but to assess how global production and ownership evolves, we adjust the production of future mines calculated with Taylor's law so that global production in 2030 matches demand projections from the International Energy Agency's Announced Pledges Scenario. We use the Announced Pledges Scenario as it is widely used in policymaking, and its associated primary metal demand has been extensively analyzed in the Global Critical Minerals Outlook 2024 (IEA, 2024).

3 Results and discussion

3.1 Historical trends of production vs. Ownership

Fig. 1 illustrates the disaggregation of copper production (first column) and copper ownership (second column) by region (first row) and by income category (second row). In the income breakdown (second row), values are presented as percentages, with the “unknown” ownership category (displayed in dark gray in Fig. 1b) excluded from

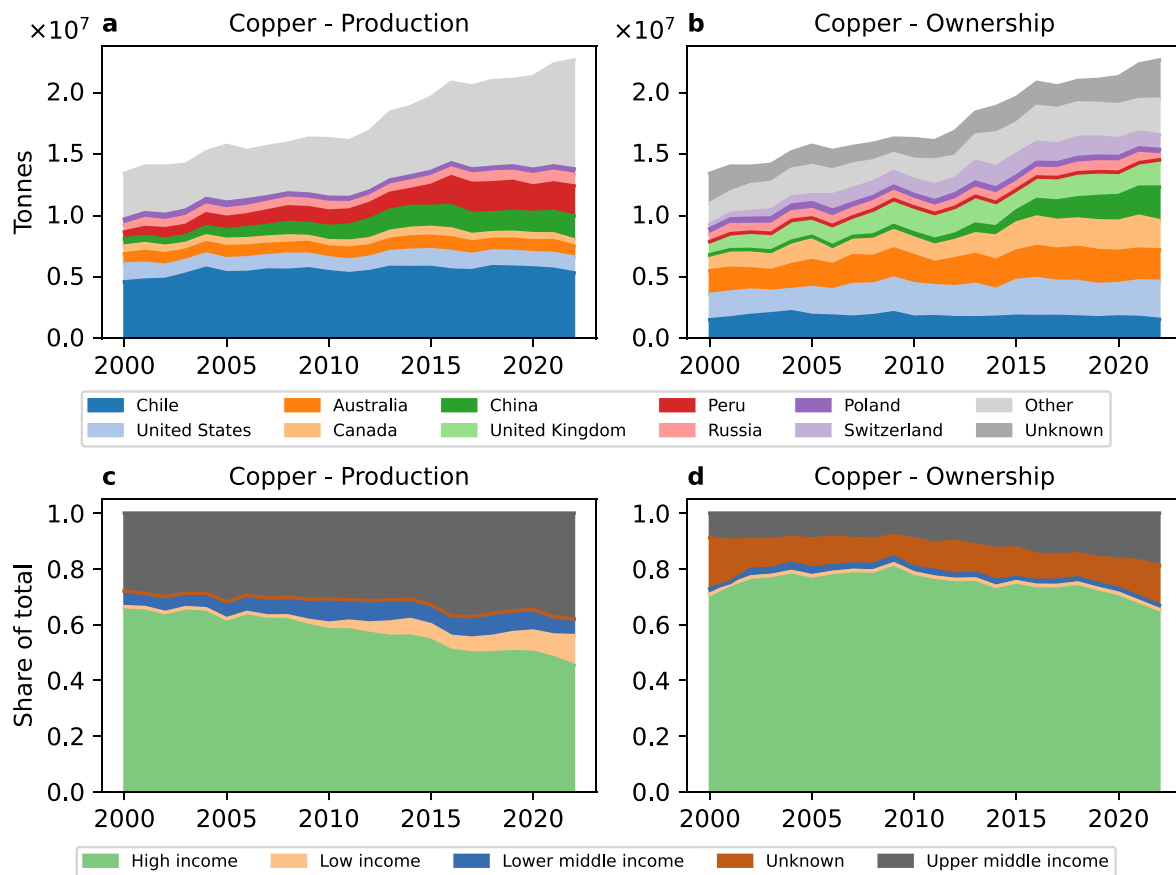


Fig. 1. Copper production and ownership. a–b. By region. c–d. By income class.

the calculation. Supplementary Figures 5–15 provide analogous results for the other metals or metal ores analyzed in this study.

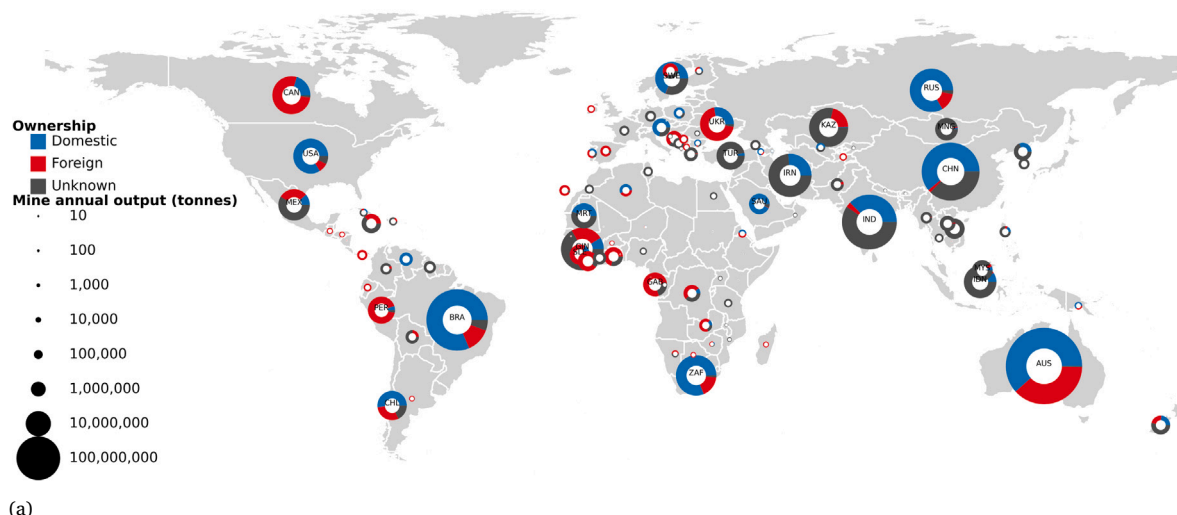
From 2000 to 2022, global mining output increased in response to rising industrial and technological demand. For example, copper production increased from 13 Mt in 2000 to 22 Mt in 2022 (Figs. 1a–b). For most metals, production in high-income countries declined over time while increasing in upper-middle-income countries, with the share of global copper produced by high-income countries dropping from 0.67 in 2000 to 0.45 in 2022 and that from upper-middle-income countries rising from 0.28 to 0.38 (Fig. 1c). As shown in Supplementary Figures 6 and 10, two notable exceptions were observed: lithium production, which saw an increased concentration in high-income countries (rising from 0.56 to 0.79), and cobalt production, which became increasingly concentrated in low-income countries (with cobalt's share in low-income regions rising from 0.21 in 2000 to 0.72 in 2022, notably driven by the Democratic Republic of Congo with a share of 0.70).

Although high-income countries continue to maintain significant ownership stakes, their share of ownership has generally declined over time, while ownership held by upper-middle-income countries has expanded. For instance, high-income ownership of uranium assets decreased from 0.62 to 0.46 between 2000 and 2022, whereas ownership by upper-middle-income countries increased from nearly 0 to 0.35 during the same period. Countries such as Indonesia (for nickel and cobalt), the DRC (for copper and cobalt), Kazakhstan (for uranium and iron ore), and Guinea (for bauxite) have experienced substantial increases in production. China consistently emerges as both a top producer (for bauxite, iron ore, copper, and zinc) and a rapidly growing owner of mines abroad, with significant acquisitions in Africa, Latin America, and Asia. Meanwhile, despite occasional declines in production in Canada and the U.S. (for copper, nickel, lead, and zinc), ownership positions in these countries remain robust or even grow (as observed, for example, in U.S. ownership of copper). Australia, by

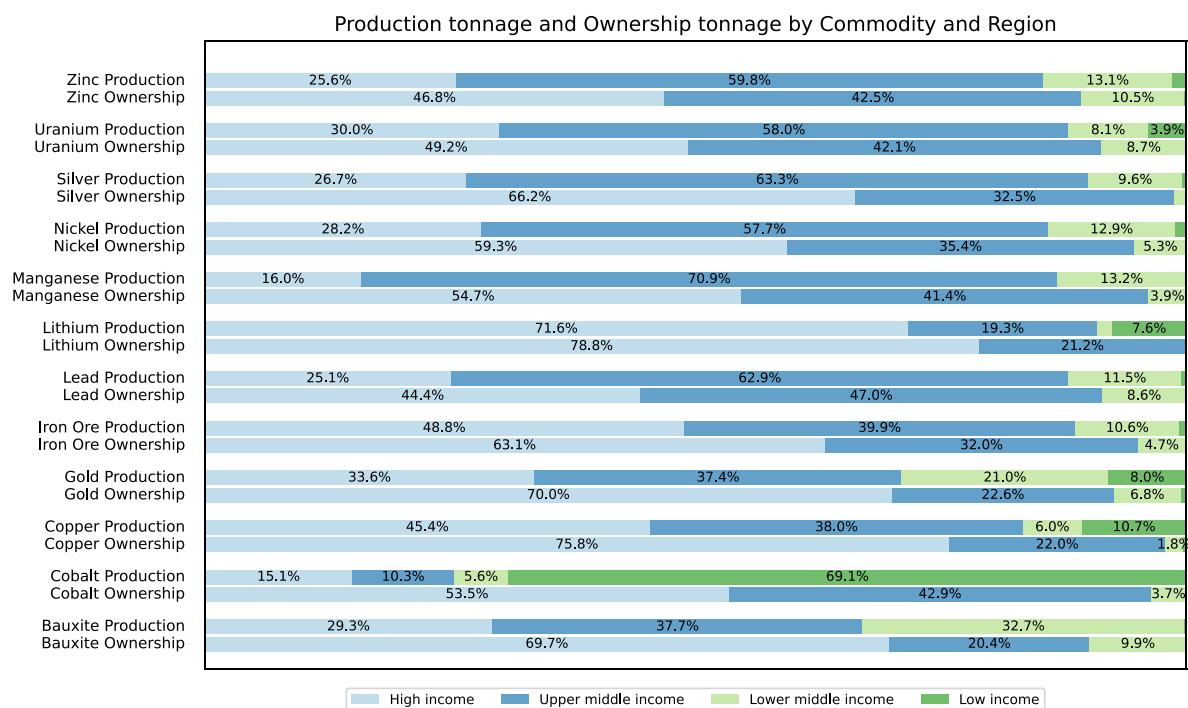
contrast, has shown consistent expansion in both production (across iron ore, bauxite, lithium, and manganese) and ownership.

Fig. 2(a) further delineates the ownership structure by presenting the breakdown of total production across all 12 metals and metal ores in 2022 controlled by domestic companies (in blue), foreign companies (in red), and unknown owners (in gray). Notably, several countries with a significant share of “unknown” production also exhibit high rates of domestic control. For example, China controls 61% of its production while 37% remains unknown; India controls 59% with 37% unknown; Iran, by contrast, controls only 26% with a high unknown share of 74%. Countries such as Turkey and Indonesia are characterized by exceptionally high percentages of production under unknown ownership (97% and 88%, respectively).

A complementary perspective is to ask what fraction of global output is governed through foreign direct investment (FDI). Using the ownership database for 2019, we calculate, for each metal, the share of mine production held by companies headquartered outside the producing country. Because a non-trivial share of production, ranging from 9% to 40% depending on the metal, cannot be matched to a known owner, we report a conservative lower bound (assuming all unknowns are domestically owned) and an upper bound (assuming they are all foreign-owned). The results show that FDI governs between 19%–34% of uranium production, 44%–53% of lithium, 57%–66% of copper, 83%–91% of cobalt, 22%–44% of iron ore, 35%–60% of manganese, 31%–60% of zinc, 24%–72% of nickel, 33%–70% of silver, 38%–70% of gold, 19%–62% of lead, and 36%–80% of bauxite. These ranges are consistent with estimates from Sun et al. (2024), who reported that 47% of lithium, 71% of cobalt, and 41% of nickel production in 2019 were under foreign control. Even under conservative assumptions, a substantial share of global mining is governed by FDI.



(a)



(b)

Fig. 2. Production vs. ownership in 2022. a. Map representing the share of production controlled either locally (in blue), by foreign companies (in red), or by unknown companies (in gray). See supplementary figure 17 for a disaggregation of the maps by commodity. b. Disaggregation of the production and the ownership by income class, excluding unknown owners (see supplementary figure 16 for the same figure including unknown owners).

Across multiple metals, a common pattern emerges: high-income countries tend to hold disproportionately large ownership stakes relative to their share of physical production (Fig. 2(b)). In 2022, high-income countries controlled 49% of uranium assets while accounting for only 30% of global uranium production. Similarly, in the case of copper, high-income entities owned 76% of assets despite producing only 45% of the global output. Conversely, upper-middle-income countries appear to be increasingly responsible for the physical production of several key metals, such as uranium, nickel, and zinc.

Our findings are consistent with the historical trajectory outlined by Ericsson (2022), which shows that while the geographic locus of mining continues to shift towards emerging economies, strategic control remains concentrated in established high-income centers. This geographic redistribution of production could exacerbate environmental

and social challenges, particularly in countries where local institutions may struggle to manage the impacts of mining. It raises important questions regarding the distribution of economic benefits and the allocation of risks. High-income nations may be insulated from many of the direct impacts of resource extraction, yet they continue to exert significant influence over the global supply of critical minerals.

Moreover, the persistent category of “unknown” ownership presents a challenge for transparency and accountability in the mining sector. The high percentage of unknown ownership in certain countries may obscure the true nature of control and hinder effective policy responses aimed at mitigating environmental and social risks. For example, metals with a known lack of supply chain transparency include gold (34% of unknown ownership in 2022, see supplementary figure 16) which is mined and then allegedly sold illegally (Greef, 2023; Dammer and Niño, 2024).

In addition to supply chain vulnerabilities, our findings invite consideration of the broader socio-environmental consequences of foreign ownership. In many low- and middle-income countries, mining is central to socioeconomic development (Ericsson and Löf, 2019). However, when high-income country-based entities own mines in emerging economies, local communities may perceive the distribution of benefits as inequitable while bearing disproportionate social and environmental burdens. This situation can lead to a “liability of foreignness”, where foreign-owned operations – often less embedded in local networks and less responsive to community needs – generate heightened social conflict and protest (Haslam et al., 2019; Berman et al., 2017). Although one study found no significant link between partial Chinese or tax haven-based ownership and protest incidence (Christensen, 2019), government involvement appears to mitigate conflict by either enforcing regulations or by selectively investing in low-risk projects that bring additional benefits and infrastructure (Christensen, 2019).

Moreover, in their efforts to attract foreign capital, governments in developing economies may offer mineral rights on very favorable terms despite significant political risks. Such low-cost concessions can substantially increase a country's resource FDI share, even when the risk of expropriation is high (Hajzler, 2014). Recent developments in Ukraine – where the US has proposed opportunistic deals that offer disadvantageous conditions in exchange for greater access to the country's critical minerals – further exemplify how such arrangements can be used to leverage resource wealth amid geopolitical pressures (Gayle, 2025). The close nexus between resource extraction and FDI can erode national institutions – particularly property rights – and compound long-term vulnerabilities (Chiyaba and Singleton, 2024). Additionally, FDI in the resource sector may lead to varying outcomes in environmental degradation, depending on the host country's regulatory capacity and income level (Muhammad et al., 2021), and it may even contribute to less stringent environmental regulations when investment inflows are high in politically weak settings (Van et al., 2024). Nonetheless, while FDI in mining is traditionally seen as resource-seeking, it may also offer the potential for technology transfer and local innovation (Casella and Formenti, 2022).

Reshaping trade mechanisms to better serve developing countries' interests will likely require proactive measures at both national and international levels. One strategic avenue is for resource-rich developing nations to capture more value domestically before export, rather than simply feeding established overseas supply chains. This involves policies to encourage local beneficiation, such as building refining and processing capacity at home. Many African countries are now exploring this approach. African leaders are adopting measures like export bans on unrefined minerals, infrastructure-for-access deals, and regional co-operative refineries, all aimed at shifting from raw export dependency to local value addition (Pratt, 2025). Such steps can gradually alter trade routes by retaining materials for processing in-country or within the region, thereby breaking the pattern of straight export of ores to traditional industrial centers. Another key measure is strengthening the negotiating leverage and governance capacity of developing countries in their dealings with foreign investors and buyers. With the right host-country policies (strong regulations, accountability and standards) foreign investment can upgrade local industry and embed new supply relationships benefiting the host (Kotschwar et al., 2012). International cooperation could further help rebalance the global trade architecture. Initiatives like an African Union-backed continental investment charter and regional processing hubs (Pratt, 2025) are designed to give resource-rich developing regions more collective clout in negotiations and more control over supply chains. By coordinating policies, sharing infrastructure, and jointly setting terms (for example, minimum local content or fair offtake agreements), developing countries can avoid a race-to-the-bottom and instead bargain for trade arrangements that prioritize their development goals.

3.2 Relationship between ownership and trade

Our analysis first quantified the practical overlap between foreign mine ownership and international ore trade. The resulting overlap shares, averaged across 2000–2022, ranged from 2.0% (iron ore) to 13.8% (lithium), indicating that at most a modest proportion of global trade value in ores and concentrates aligns directly with existing patterns of foreign ownership (Table 1). This descriptive measure provides an upper bound on the potential influence of foreign corporate control on ore trade. We then examined whether the presence of foreign ownership increased the probability of a trade connection using logistic regression with year fixed-effects. Despite large estimated odds ratios, none were statistically significant (all p-values > 0.93), indicating no evidence that foreign ownership increases the likelihood of observing bilateral trade relationships. Finally, we assessed trade intensity conditional on both ownership and trade links by estimating two-way fixed-effects regressions of log trade value on log owned production. Exporter, importer, and year intercepts absorbed nearly all within-dyad variation, rendering elasticity estimates unidentifiable for most metals. Copper ($\beta = -0.15$, $p = 0.45$) and gold ($\beta = 0.33$, $p = 0.27$) were the only exceptions with valid standard errors, and neither coefficient was statistically significant.

When we applied the PPML regression that focuses on year-to-year changes within each country pair, the estimator did not converge to a finite coefficient for most metals. In practical terms, non-convergence means the model cannot find a stable average relationship between changes in foreign ownership and subsequent changes in trade once exporter-by-year and importer-by-year conditions are held fixed. This occurs here because most country pairs either never trade at all or, when ownership changes, their trade values remain zero; after accounting for conditions on each side, the remaining variation is too limited and too sparse to identify a common effect. In the few cases where the model did converge, the estimated effects of lagged ownership on trade were small and statistically indistinguishable from zero. Taken together, these results reinforce our main conclusion: at the macro level, foreign mine ownership is not a reliable predictor of bilateral trade in ores and concentrates. Trade patterns appear to be governed primarily by other factors, which could be long-term contracts, midstream processing locations, logistics, and policy.

Trade is therefore influenced by other factors than ownership. FDI in natural resources typically involves equity stakes, loan-for-offtake agreements, and long-term contracts, meaning that controlling ownership may aim to diversify supply or expand production rather than secure exclusive output (Kotschwar et al., 2012). Varying investment motivations, such as market-seeking versus export platform strategies, can differently impact trade performance (Franco, 2013) and could contribute to the observed decoupling. Moreover, as in the oil industry (Bridge, 2008), global raw material networks are shaped more by inter-firm relationships than by state-to-state ties. Legal regimes and trade policies, including export restrictions and preferential trade agreements, further complicate the link between production and corporate control (Srivastava, 2023).

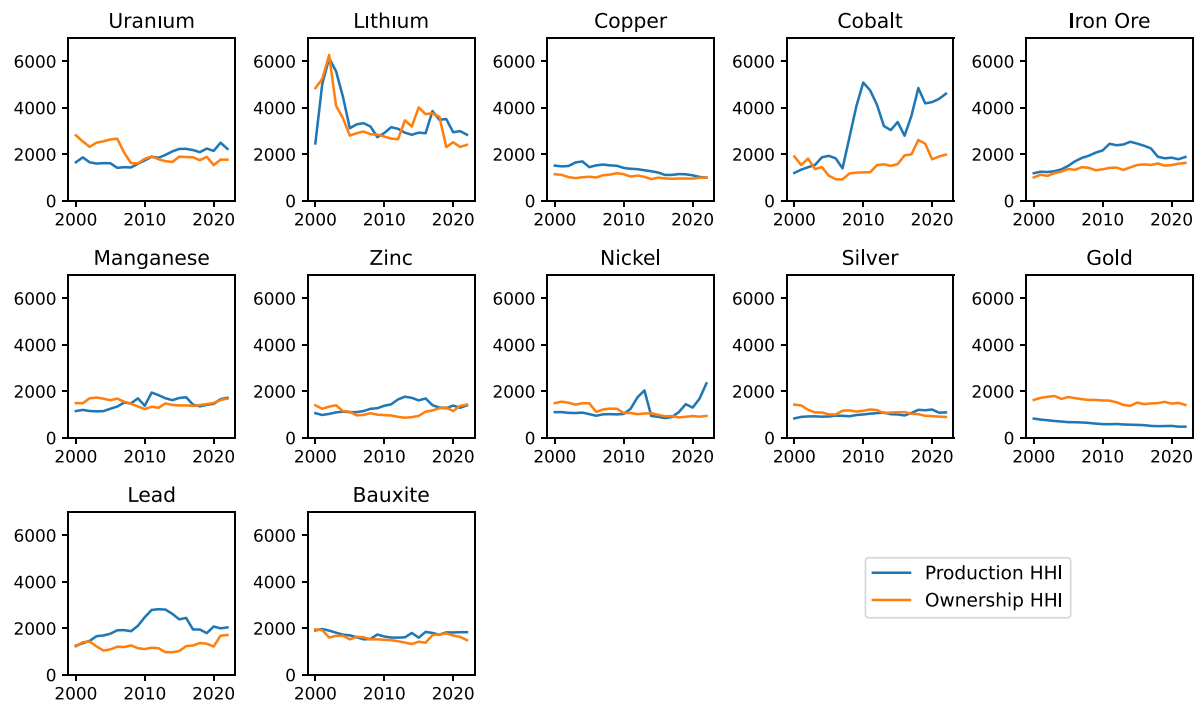
Micro-level studies demonstrate that foreign ownership not only facilitates increased exports to the investor's home country but also enables firms to leverage their foreign status to access broader international markets (Li et al., 2024; Boddin et al., 2017). Thus, the misalignment between trade and ownership flows in our analysis may reflect strategic diversification rather than an absence of influence. Additional insights from bilateral relations literature indicate that factors like political connections (Ding et al., 2018; Sharma et al., 2020), bilateral trust (Guiso et al., 2009), linguistic proximity (Melitz and Toubal, 2014), and both formal and informal institutional ties (Haveman et al., 2017; Araujo et al., 2016; Li et al., 2023) also significantly enhance trade flows.

Our finding of no statistical correlation between mine ownership and trade flows does not diminish the relevance of corporate control as

Table 1

Overlap, incidence, and intensity results for the relationship between foreign mine ownership and ore trade flows.

Metal	Overlap share (%)	Incidence (logit)		Intensity (log-log FE)		Observations
	OverlapShare	Odds ratio	p-value	Elasticity	p-value	
Bauxite	9.95	1.97×10^{12}	0.99	1.60	–	201
Cobalt	9.60	9.01×10^9	0.93	–1.15	–	48
Copper	4.20	6.37×10^{12}	0.99	–0.15	0.45	838
Gold	3.65	9.14×10^{12}	1.00	0.33	0.27	780
Iron Ore	2.01	2.35×10^{11}	0.97	0.13	–	360
Lead	8.61	9.02×10^{11}	0.98	0.49	–	278
Lithium	13.76	1.02×10^{12}	0.99	1.47	–	61
Manganese	3.51	1.30×10^{12}	0.99	–0.23	–	98
Nickel	4.77	2.86×10^{10}	0.94	–0.54	–	147
Silver	12.65	2.25×10^{12}	0.99	0.24	–	378
Uranium	11.16	8.59×10^{10}	0.96	–0.32	–	95
Zinc	12.83	5.82×10^{12}	0.99	0.21	–	461

**Fig. 3.** Herfindahl–Hirschman Index (HHI) comparisons for production and ownership.

a factor in supply-chain security. Corporate control represents who ultimately decides how and where resources are developed. Even if trade patterns today are driven more by market logistics than ownership ties, companies controlling mines have the power to influence supply in other ways. For instance, mining firms have been known to intentionally curtail or redirect production for strategic advantage (Lèbre et al., 2021). In a scenario where a few large firms dominate a metal's production, those firms could become potential gatekeepers of the material. They have no interest in doing so in the functioning market today but it is nonetheless a point of leverage. For example, if such companies were to act in concert or under state direction, they might reroute or restrict exports for political or economic reasons, thus directly impacting global supply chains. Criticality assessments have both a vulnerability and a probability dimension. Our results show that the probability of owners disrupting the supply chain is low. However, as the vulnerability to such disruptions is likely to be high, we argue that corporate ownership should nonetheless be integrated into criticality assessments.

Fig. 3 reports the results from the Herfindahl–Hirschman Index (HHI) analysis. In 2022, cobalt's production HHI reached 4602 due to the DRC's dominance in global output, while its ownership concentration (HHI = 1985) indicates that although extraction is geographically concentrated in the DRC, control is held by a diverse set of international

companies. Nickel production has become highly concentrated (HHI 2340 in 2022), driven by Indonesia's growing share, while its ownership concentration stayed constant. For lithium, the HHI increased from 2464 to 2839 in production, mainly due to growing output from Australia, Chile, and China; however, ownership concentration declined from a highly concentrated HHI of 4842 in 2000 to a moderate level of 2407 in 2022, highlighting broader corporate participation. In contrast, base and bulk metals and metal ores such as copper (HHI ≈ 1000 for both ownership and production), silver (approximately 1100 for production and 900 for ownership), and bauxite (HHI around 1800 for production and 1500 for ownership) remain more widely distributed across producers. Uranium exhibits an inverse trend, with rising production concentration (HHI of 2231 in 2022) as Kazakhstan gains dominance, while ownership concentration decreases (HHI of 1767 in 2022), indicative of diversified investments. Gold shows a slight decrease over time in both production and ownership HHIs, with ownership showing higher concentration (being the only commodity studied to have consistently higher ownership concentration).

Although the U.S. Department of Justice and the Federal Trade Commission (2010) offer HHI guidelines – categorizing markets as unconcentrated (HHI < 1500), moderately concentrated (HHI 1500–2500), or highly concentrated (HHI > 2500) – recent research challenges the use

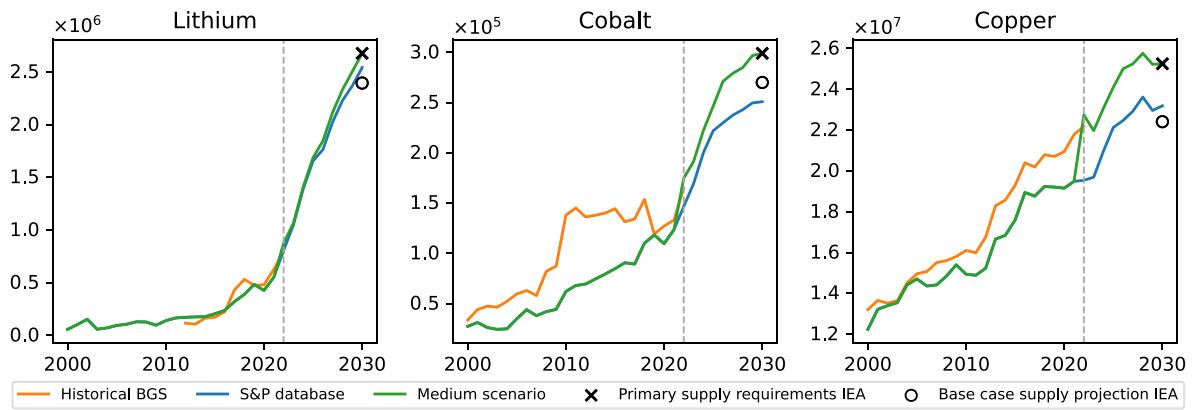


Fig. 4. Future production projections up to 2030 aligned with IEA demand projections. Lithium is in Lithium carbonate equivalent, while Cobalt and Copper are in metric tonnes of metal content.

of fixed thresholds, arguing that such benchmarks may underestimate supply risks in less concentrated markets (Bucciarelli et al., 2025). Nonetheless, significant disparities in market concentration across metals exist. The high concentration observed for cobalt, and lithium underscores potential supply chain vulnerabilities, as a few dominant countries control a large share of production. High production concentration in certain regions may expose the supply chain to risks arising from geopolitical instability, resource nationalism, or trade restrictions. While corporate concentration at the mine stage may not immediately trigger market failures, high concentration at the refinery stage – largely driven by centralized state control – poses serious supply security risks (Ericsson et al., 2024a). Concentration levels vary significantly along the value chain, with extraction-stage HHI declining for some metals while midstream processes, such as smelting and refining, become increasingly concentrated, particularly due to strategic moves by dominant players like China (Bucciarelli et al., 2025).

For completeness we also calculated HHIs at the mine scale for production and at the immediate-owner scale for corporate control (Supplementary Fig. 18). As expected, mine-level HHIs are markedly lower than the country-level figures discussed above (e.g. copper: 292 in 2000 and 132 in 2022), because the index is now distributed across several hundred individual sites rather than a few dozen producing states. By contrast, ownership computed for the named equity holders is slightly more concentrated than mine production (for most metals the 2022 owner-level HHI is 1.1–1.7 times the corresponding mine-level value), reflecting the fact that a single company typically controls several mines. These company-scale results must, however, be interpreted with caution. The S&P database lists owners exactly as reported in regulatory filings, which means that subsidiaries, joint-venture vehicles, or special-purpose entities often appear as distinct firms even when they are ultimately controlled by the same parent group. Without systematic consolidation of such affiliations, the number of “firms” is overstated and the ownership HHI is biased downwards, while the splitting of production across nominally separate entities can also bias it upwards relative to the mine-level benchmark. Because the direction and magnitude of these two effects vary by jurisdiction and by commodity, the company-level HHIs presented here should be considered indicative only. A more reliable appraisal of corporate concentration would require explicit hierarchy matching or ultimate-parent reconciliation, something that lies beyond the scope of the present study but represents an important avenue for future work.

3.3 Future production and ownership

Our future production scenario (shown as the green curve in Fig. 4) aligns with the IEA’s projected 2030 primary supply demand (indicated by the black cross). In contrast, the blue curve represents the S&P database baseline (i.e., without any additional assumptions). The gap

between the green and blue curves represents the projects for which supplementary assumptions were made. The circular marker on Fig. 4 for the year 2030 corresponds to the IEA’s “base case supply projection”, which, similar to the blue curve, does not account for additional new projects.

For copper, the production distribution remains largely stable. Ownership shifts are minor except for Canada who’s ownership share rises from 11% to 17%. Lithium undergoes greater shifts. Australia’s production share drops from 46% to 27%, while the U.S. rises from 2% to 6% and Chile falls from 25% to 12%. Ownership changes include a modest increase for Australia (27% to 29%) and a sharp rise for Canada (1% to 12%); meanwhile, U.S. ownership declines (19% to 7%). Cobalt also sees a redistribution: the DRC’s production share falls from 71% to 45%, while Australia (4% to 8%) and Canada (3% to 5%) expand. Ownership follows a similar pattern: Australian control rises (3% to 10%), Canada’s increases (2% to 8%) (see Fig. 5 and supplementary figure 19 for the percentages).

These projections rest on the simplifying assumption that the ownership of mines and mining projects observed in 2022 remain unchanged until 2030—a scenario that is unlikely to hold in practice. For example, current activities may lead to growing ownership from new high-income countries such as the United Arab Emirates in low-income countries (Bakr, 2024). The evolving geopolitical landscape is expected to significantly influence future investment flows and policy interventions (Humphreys, 2013). For instance, as observed in OECD countries (Sun et al., 2024), while advanced financial systems enable effective exploitation of mineral resources, heavy reliance on external FDI can also heighten vulnerability to commodity price fluctuations and an overdependence on finite assets. Although our analysis does not incorporate external factors such as changing regulatory frameworks or shifts in FDI, our baseline scenario provides a useful reference point. In the absence of complementary policy measures, the evolving patterns in production and ownership may indirectly amplify long-term supply vulnerabilities.

Moreover, supply risk is not limited solely to concerns around market concentration of production and ownership. It arises in combination with other issues such as environmental degradation, social inequities, economic tensions and geopolitical instability. Proactive policy interventions and transparent, multi-stakeholder governance mechanisms will be essential to mitigate these risks (Dou et al., 2023). The projected shifts in production and ownership should thus also be viewed as portents of future complications linked to environmental challenges, thereby informing policies aimed at sustainable resource management (Muhammad et al., 2021).

Future production and ownership scenarios must also be interpreted in the context of potential policy shifts. Changes in governmental policies – such as adjustments to production quotas, local ownership regulations, and other regulatory frameworks – can rapidly reshape

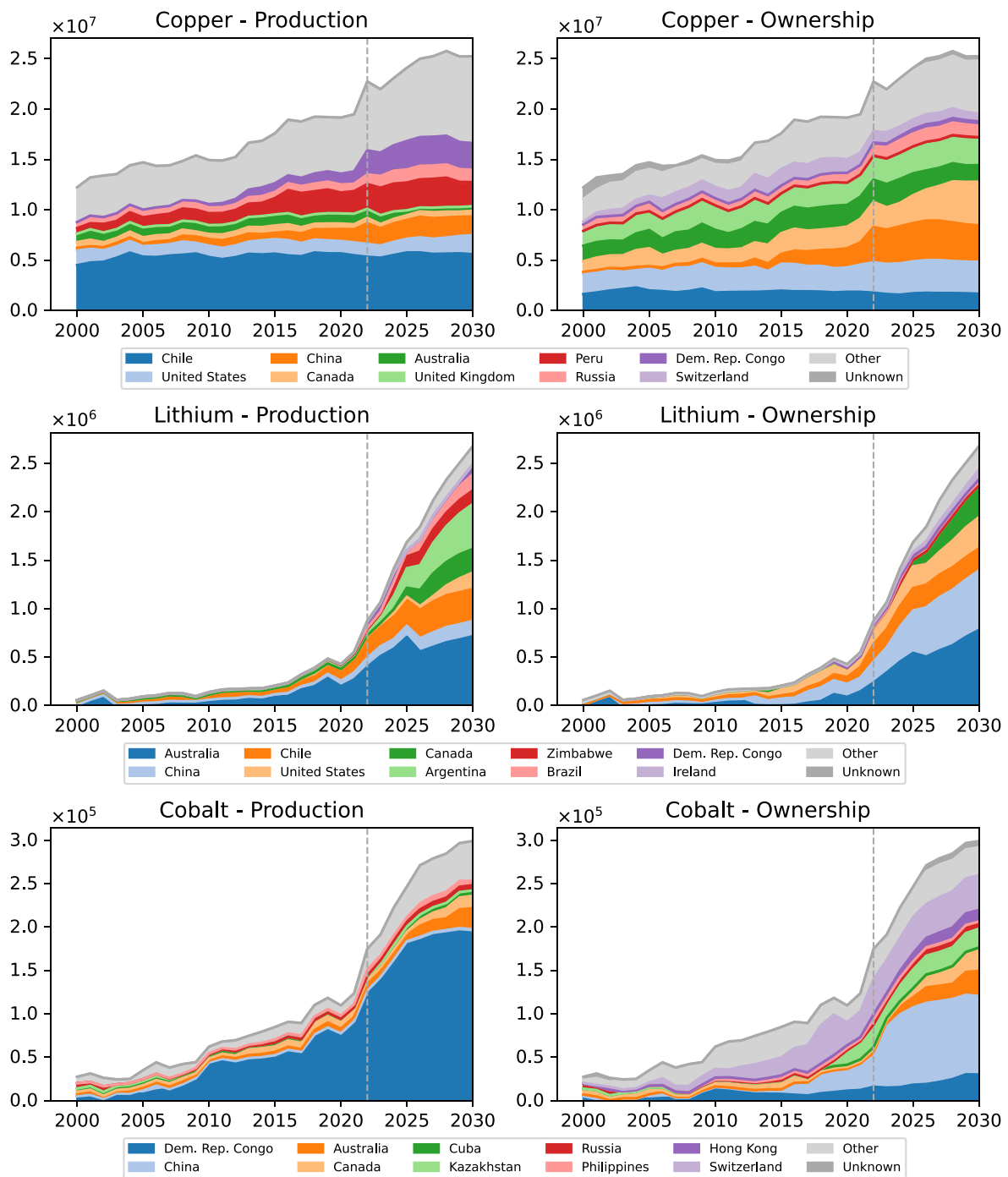


Fig. 5. Future scenarios of production and ownership by region. Lithium is in Lithium carbonate equivalent, while Cobalt and Copper are in metric tonnes of metal content.

both the supply structure and the distribution of ownership (Shen et al., 2020). Strategic policy interventions aimed at resource governance and environmental protection are likely to further influence these dynamics. For example, export tariffs, which serve not only as instruments of trade policy but also as signals of a country's commitment to resource conservation and environmental protection (Chen and Zheng, 2019), can significantly alter market dynamics by affecting price stability and supply security.

Finally, factors such as political stability, legal frameworks and enforcement, and local infrastructure are as crucial as resource endowment in shaping investment decisions (Wang et al., 2021). Together, these considerations underscore the importance of integrating market

dynamics with strategic policy measures to enhance both supply chain resilience and sustainable resource management in the future.

4 Conclusion

This paper presents analysis on the impacts of mine ownership on trade of 12 metals and metal ores. We also create estimates of future mine outputs that align with an IEA scenario. While extraction increasingly occurs in emerging economies, ownership remains concentrated in high-income countries. This duality, reflected in contrasting HHI trends, underscores supply vulnerabilities beyond conventional import concentration metrics. However, we find no correlation between

ownership and trade flows, indicating that trade relationships are shaped by factors other than corporate control. Our forward-looking scenarios for lithium, copper, and cobalt suggest that production and ownership patterns will continue evolving by 2030. Although these projections assume the ownership structure remains unchanged from 2022 onward, they provide a baseline for anticipating shifts in global resource governance. Strengthening regulatory frameworks, enhancing transparency, and integrating corporate control into policy measures will be essential for improving supply chain resilience. Overall, these findings emphasize the need to account for both production geography and corporate ownership in criticality assessments to fully capture supply risks and opportunities.

CRedit authorship contribution statement

Baptiste Andrieu: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Karla Cervantes-Barrón:** Writing – review & editing, Methodology, Conceptualization. **Raghav Pant:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Samira Barzin:** Writing – review & editing, Methodology, Conceptualization. **Mehrnoosh Heydari:** Writing – review & editing, Methodology, Conceptualization. **Gaël Parpan:** Writing – review & editing, Methodology, Conceptualization. **Jonathan Cullen:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.resconrec.2025.108647>.

Data availability

We used proprietary data (S&P Capital IQ Pro) so do not have permission to share it. However, we attached a supplementary data file which gives all our results at the country-aggregated level.

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